

ComicCraft – AI Comic Story Creator Using Gemini Models

4. Project Planning Phase

Organizing Milestones, Activities, Dependencies & Deliverables

1. Planning Objective

The project guide divides the implementation work into a series of milestones. This phase arranges those milestones into a practical sequence that can be followed to develop, integrate, and verify the ComicCraft application.

2. Milestone 1 – Model Selection and Architecture

Activities: Study and select suitable generative AI models, define the overall application architecture, and prepare the development environment. **Deliverables:** Defined model responsibilities, an architecture outline, and a ready-to-use project environment.

3. Milestone 2 – Core Functionality Development

Activities: Develop outline generation, story generation, image generation, comic layout construction, and PDF export. Implement the FastAPI backend for routing and processing user inputs. **Deliverables:** Core generation modules together with backend integration.

4. Milestone 3 – routes.py Development

Activity: Develop the main application logic and route handlers inside **routes.py**. **Deliverable:** Application routes that connect user requests with the comic-generation workflow and provide failure handling.

5. Milestone 4 – Frontend Development

Activities: Design and build the user interface and create dynamic templates using FastAPI's Jinja2 integration. **Deliverables:** Homepage, comic preview page, and export-success page.

6. Milestone 5 – Deployment and Local Verification

Activities: Prepare the application for local deployment, execute it locally, and verify its operation. **Deliverables:** Configured development environment, running local server, and verified browser/API access.

7. Dependencies and Sequence

The model and architecture decisions provide the basis for implementing the application modules. After the core functions are developed, they are connected through the routes. The frontend templates then present the results generated by the backend. Local deployment and verification are performed after the application components have been integrated.

8. Planning Considerations

The development plan should include preparation of the Python environment and required dependencies, configuration of Gemini and Hugging Face credentials through environment variables, and confirmation that the static and template directories are available. Sufficient time should also be allocated to test both browser-based and API-based workflows.

9. Planning Outcome

The project plan establishes a milestone-driven path for ComicCraft, beginning with initial setup and model selection and continuing through core functionality, route integration, frontend development, deployment, and local verification. This sequence helps organize the development activities and their related deliverables.